

Divisibility rules, remainders, prime factors.
Equations with integer solutions.

1. How many four digit multiples of 9 consist only of four different odd digits

- A 4 B 12 C 24 D 36 E 120

1 3 5 7 9 $\Sigma = 25$ 4 digits $4 \times 3 \times 2 \times 1 = 24$
X

2. How many three digit numbers are divisible by 5 or 9?

- A 200 B 220 C 240 D 260 E 280

100 - 999 900 numbers

$$\frac{900}{5} = 180 \text{ numbers divisible by } 5$$

$$\frac{900}{9} = 100 \text{ numbers divisible by } 9$$

$$\frac{900}{45} = 20 \text{ numbers divisible by } 45$$

$$180 + 100 - 20 = 260$$

3. The digit p, q, r, s, t are all different.

What is the smallest five digit integer ' $pqrst$ ' that is divisible by 1, 2, 3, 4 and 5?

Last digit 0 Multiple of 10

Consider smallest $123 \underline{\quad} 0$
 $\uparrow$
 6

12360

4. Find the possible values of the digits p and q, given that the five digit number 'p543q' is a multiple of 45.

multiple of $5 + 9$ $q = 0, 5$

$$q = 0 \quad 6 + 4 + 3 + 0$$

$$q_r = 5 \quad 5 + 4 + 3 + 5$$

$$p = 6$$

$$p = 6$$

$$\rho = 1$$

so sum is multiple of 9

5. $30^{20} - 20^{20}$ is divisible by all the following values except for which one

- A 10 B 25 C 40 D 60 E 64

$$= (2 \times 3 \times 5)^{20} - (2 \times 2 \times 5)^{20}$$

$$= 2^{20} \times 5^{20} [3^{20} - 2^{20}]$$

6. If n is a multiple of 5 and $n = p^2q$, where p and q are prime numbers, which of the following must be a multiple of 25?

A p^2 B q^2 C pq D p^2q^2 E p^3q

either p or q must be a multiple of 5
so p^2q^2 must be a multiple of 25

7. If both 11^2 and 3^3 are factors of the number $a \times 4^3 \times 6^2 \times 13^9$, what is the smallest value of a ?

A 33 B 121 C 363 D 1089 E 3267

$$11^2 \times 3 = 121 \times 3 = 363$$

8. What is the highest positive integer power of 3, that divides $100!$?

($100! = 100 \times 99 \times 98 \times \dots \times 3 \times 2 \times 1$)

A 33 B 48 C 60 D 99 E 100

$$\frac{100}{3} = 33 \text{ multiples of } 3$$

$$\frac{100}{27} = 3 \text{ multiples of } 3^3 \quad 33 + 11 + 3 + 1$$

$$\frac{100}{9} = 11 \text{ multiples of } 3^2$$

$$\frac{100}{81} = 1 \text{ multiple of } 3^4$$

9. What is the remainder when $|5^{13} - 6^{14}|$ is divided by 10?

A 1 B 2 C 5 D 6 E 9

6^{14} last digit = 6
 5^{13} last digit = 5

$$6 - 5 = 1$$

10. How many different positive integers exist between 10^6 and 10^7 , the sum of whose digits is equal to 2?

A 6 B 7 C 8 D 12 E 18

$$1000000 \rightarrow 10,000,000$$

7 digits 1, 1, 50's or 2, 60's

1st digit 1

6 options for '1'

1st digit 2

all 0's

$$6 + 1 = 7$$

11. An integer n is divided by an integer k and leaves a remainder of 24.
When twice the integer n is divided by k the remainder is 11.
What is the value of k ?

A 13 B 35 C 37 D 59 E 18

$$\begin{aligned} n &= ak + 24 & 2n &= bk + 11 \\ 2n &= 2ak + 48 & (b-2a)k &= 37 \\ 2ak + 48 &= bk + 11 & \text{integer} \times \text{integer} &= 37 \end{aligned}$$

1, 37

12. What is the largest prime factor of $7^{20} + 7^{21} + 7^{22}$

A 7 B 13 C 19 D 57 E 7^{20}

$$7^{20}(1 + 7 + 49) = 7^{20} \times 57 = 7^{20} \times 3 \times 19$$

13. A natural number k is such that $k^2 < 2023 < (k+1)^2$
What is the largest prime factor of k

A 5 B 7 C 11 D 13 E 19

$$40^2 < 2023 < 50^2$$

$$\begin{array}{r} 45 \\ 45 \\ \hline 225 \\ 1800 \\ \hline 2025 \end{array}$$

$$44^2 < 2023 < 45^2$$

$$44 = 2 \times 2 \times 11$$

14. The positive integer k is the smallest value such that $140k$ is a perfect square.
What is the sum of the digits of k ?

A 5 B 8 C 11 D 14 E 15

$$140k = 2 \times 2 \times 5 \times 7 \times k$$

$$k = 5 \times 7 = 35$$

15. How many positive integers that are factors of $(2^3)(3)(5^2)$ are even?

A 6 B 12 C 16 D 18 E 24

1:	2	4	8
3:	6	12	24
5:	10	20	40
3×5 :	30	60	120
5×5 :	50	100	200
$3 \times 5 \times 5$:	150	300	600

Equations with integer solutions

even : $2n$ is factor 2, 6, 10, 30, ...

$$75 = 3 \times 5 \times 5$$

odd : factor of 75 3, 5, 15, 25,

1. How many ways can 75 be expressed as the sum of two or more consecutive positive integers?

A 1 B 2 C 3 D 4 E 5

2: $37 + 38$

6: 10 11 12 13 14 15

10: ... 7 8 ...

3: 24 25 26

15: ... 5 ...

← could be -ve

5: 13 14 15 16 17

2. Consider a sequence of five consecutive positive integers. How many such sequences are there, in which the sum of the squares of the first three integers is equal to the sum of the squares of the other two integers?

A 0 B 1 C 2 D 3 E 4

$$(n-2)^2 + (n-1)^2 + n^2 = (n+1)^2 + (n+2)^2$$

$$n^2 - 4n + 4 + n^2 - 2n + 1 + n^2 = n^2 + 2n + 1 + n^2 + 4n + 4$$

$$n^2 - 12n = 0 \quad n = 0, n = 12$$

3. Given the equation $xy = 8x + 8y - 16$, where x and y are positive integers, find the smallest possible value of xy

A 0 B 16 C 64 D 224 E 240

$$xy - 8x - 8y = -16$$

$$(y-8)(x-8) - 64 = -16$$

$$(y-8)(x-8) = 48$$

$$\begin{array}{l} = 1 \times 48 \\ \quad 2 \times 24 \\ \quad 3 \times 16 \end{array} \quad \begin{array}{l} 4 \times 12 \\ 6 \times 8 \end{array}$$

$$\begin{array}{r} 16 \\ 14 \\ \hline 64 \\ 160 \\ \hline 224 \end{array}$$

4. How many ways are there to express $\frac{2}{15}$ in the form $\frac{1}{a} + \frac{1}{b}$ where a, b are positive integers with $a \leq b$

A 1 B 2 C 3 D 4 E 5

$$\frac{2}{15} = \frac{a+b}{ab}$$

$$2ab = 15a + 15b$$

$$2ab - 15a - 15b = 0$$

$$4ab - 30a - 30b = 0$$

$$(2a-15)(2b-15) = 225 = 3 \times 3 \times 5 \times 5$$

$$2a-15=1$$

$$a=8$$

$$2b-15=225$$

$$b=120$$

$$\begin{array}{l} 1, 225 \\ 3, 75 \\ 5, 45 \\ 9, 25 \\ 15, 15 \end{array}$$

5. How many ways are there to express $\frac{5}{11}$ in the form $\frac{1}{x} + \frac{1}{y}$ where x, y are positive integers with $x \leq y$

A 0 B 1 C 2 D 3 E infinite

$$\frac{x+y}{xy} = \frac{5}{11}$$

$$11x + 11y = 5xy$$

$$25xy - 55x - 55y = 0$$

$$(5x-11)(5y-11) = 121$$

$$\begin{array}{r|rr} 5x-11 & 1 & 11 \\ \hline 5y-11 & 121 & 11 \end{array}$$

no integer solutions

6. How many pairs of positive integers satisfy the equation $5a - ab = 9b^2$

A 0 B 1 **C 2** D 3 E 4

$$5a - ab = 9b^2 > 0 \quad b=1 \quad 4a=9 \quad a = \frac{9}{4} \times$$

$$5a - ab > 0 \quad b=2 \quad 3a=36 \quad a=12 \quad \checkmark$$

$$a(5-b) > 0 \quad b=3 \quad 2a=81 \quad a=8\frac{1}{2} \times$$

$$a > 0 \quad 5-b > 0 \quad b=4 \quad a=144 \quad \checkmark$$

$$b < 5$$

7. How many pairs of integers satisfy the equation $(|a| - 3)(|b| + 4) = 12$ where a, b are integers

A 3 B 6 **C 10** D 12 E 24

1×12	$ a - 3 = 1$	$ a - 3 = 2$	$ a - 3 = 3$	a, b
2×6	$ a = 4$	$a = \pm 5$	$a = \pm 6$	4 ± 8
3×4	$a = \pm 4$	$ b + 4 = 6$	$ b + 4 = 4$	-4 ± 8
	$ b + 4 = 12$	$b = \pm 2$	$b = 0$	5 ± 2
	$b = \pm 8$			-5 ± 2
				$\pm 6 \quad 0$

8. How many pairs of positive integers satisfy the equation $3x + 2y = 120$, where $x < y$

A 4 B 5 **C 11** D 12 E 24

$$y = \frac{120 - 3x}{2}$$

$$= 60 - \frac{3x}{2}$$

$$\Rightarrow x \text{ even} \quad y < 60$$

x	y
2	57
4	54
6	51
...	...
38	3

+2 each time -3 each time

But $x < y$

x	y
20	30
22	27
24	24

← max x

9. How many ways are there to express $\frac{1}{12}$ in the form $\frac{1}{m} + \frac{4}{n}$

A 0 B 1 C 2 **D 3** E 4

$$\frac{n + 4m}{mn} = \frac{1}{12}$$

$$12n + 48m = mn$$

$$mn - 48m - 12n = 0$$

$$(n - 48)(m - 12) = 48 \times 12$$

$$(n - 48)(m - 12) = 2^4 \times 3 \times 2^2 \times 3 = 2^6 \times 3^2$$

n odd $\Rightarrow m$ even

$n - 48$ is odd factor = 1, 3, 9

$n = 49, 51, 57$

10. How many positive integer values of a are there such that $\frac{a + 220}{a + 4}$ is an integer

A 10 **B 12** C 14 D 20 E 22

$$= 1 + \frac{216}{a + 4} \quad 216 = 2 \times 2 \times 2 \times 3 \times 3 \times 3$$

$$a + 4 = 6, 8, 9, 12, 18, 24, 27, 36, 54, 72, 108, 216$$

12

11. How many integer values of a are there such that $\frac{a}{1011-a}$ is an even integer

A 1 B 4 **C 8** D 10 E 12

$$\frac{a}{1011-a} = k$$

$$ak + a = 1011k$$

$$a(1+k) = 1011k$$

↑ even odd ↑ even

$1011 = 3 \times 337$
 $(1+k)$ must divide 1011 as $k, k+1$ are coprime
 $k+1 = 1, 3, 337, 1011$
 $k = -1, -3, -337, -1011$ } 8 ways

$$a = \frac{1011 \times 2}{3} = 337 \times 2 = 674$$

$$a = \frac{1011 \times 336}{337} = 3 \times 336 = 1008$$

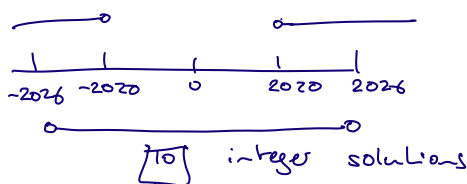
etc...

12. How many integer solutions are there to the equation $|(|x| - 2023)| < 3$

A 10 B 14 C 101 D 2020 E 2026

$$-3 < |x| - 2023 < 3$$

$$2020 < |x| < 2026$$



13. What is the greatest possible perimeter of a right-angled triangle with integer side lengths if one of the sides has length 12.

A 15 B 20 C 60 **D 84** E 120

12 is shortest side

$$12^2 + a^2 = b^2$$

$$b^2 - a^2 = 12^2$$

$$(b-a)(b+a) = 12^2$$

a, b both even or both odd

Max $a+b = 72$
 $-a+b = 12$
 $b = 42$
 $a = 36$

14. If $3^x + 2^y = 985$ and $3^x - 2^y = 473$, what is the value of xy

A 40 B 45 **C 48** D 54 E 60

$$2 \cdot 2^y = 512 = 2^9$$

$$y+1=9 \quad y=8$$

$$3^x = 985 - 256 = 729$$

$$x=6$$

$$xy = 48$$

15. What is the largest integer n such that $\frac{50!}{(5!)^n}$ is an integer

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 5 \times 2^2 \times 3 \times 2 = 5 \times 3 \times 2^3$$

$$(5!)^n = 5^n \times 3^n \times 2^{3n}$$

A 5

B 12

C 15

D 20

E 25

Need $5^n \mid 50!$ $\frac{50}{5} + \frac{50}{25} = 12$ $n \leq 12$

$3^n \mid 50!$ $\frac{50}{3} + \frac{50}{9} = 17 + 5 = 22$ $n \leq 22$

$2^{3n} \mid 50!$ $\frac{50}{2} + \frac{50}{4} + \frac{50}{8} + \frac{50}{16} + \frac{50}{32} = 25 + 12 + 6 + 3 + 1 = 47$ $3n \leq 47$
 $n \leq 15$

16. What is the sum of all the possible integers n such that $n^2 + 6n + 24$ is a square number

A -12

B -6

C 0

D 4

E 20

$$n^2 + 6n + 24 = k^2$$

$$n^2 + 6n + 24 - k^2 = 0$$

$$n = \frac{-6 \pm \sqrt{36 + 4(k^2 - 24)}}{2}$$

$$n = -3 \pm \sqrt{k^2 - 15}$$

need $k^2 - 15 = a^2$
 $k^2 - a^2 = 15$
 $(k-a)(k+a) = 15$

or $(n+3)^2 + 15 = k^2$
 $a = n+3$
 $k^2 - a^2 = 15$
 $(k-a)(k+a) = 15$

$k-a$	1	15	-1	-15	3	5	-3	-5
$k+a$	15	1	-15	-1	5	3	-5	-3
k	8	8	-8	-8	4	4	-4	-4
a	7	-7	-7	7	1	-1	-1	1
n	4	-10			-2	-4		